

Marco Mo - Data Scientist

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SUMMARY

Data Scientist applying predictive modeling, NLP, time-series analysis, and rigorous validation. Builds reproducible Python and SQL workflows and communicates findings through diagnostics, dashboards, and stakeholder recommendations.

EXPERIENCE

Junior AI Specialist

Jun 2026 - Present

ADC Nederland | Eindhoven, The Netherlands

- Developed four internal AI and data systems using Python, SQL, and typed APIs.
- Analyzed 7,773 files, reporting 6,044 conversions, duplicates, failures, and lookup gaps.
- Built process-mining KPIs for delays, rework, workload, and production bottlenecks.
- Translated operational findings into dashboards and recommendations for department directors.

Teaching Assistant

Sep 2021 - Jul 2025

Eindhoven University of Technology | Eindhoven, The Netherlands

- Supported 80+ students weekly in Python, pandas, SQL, and machine learning.
- Diagnosed modeling, coding, and interpretation errors through structured analytical coaching.
- Evaluated assignments consistently and communicated actionable feedback on methods and results.

Software Technical Lead

Sep 2021 - Jul 2023

Student Team SimEnergy | Eindhoven University of Technology

- Led six-person team building time-series models for appliance-level energy disaggregation.
- Presented prototype integrating sensor data, predictions, aggregation, and interactive visualization.

PROJECTS

Fraud Detection MLOps Pipeline - IEEE-CIS

- Built predictive modeling pipeline for 600,000 transactions and 400+ features.
- Prevented temporal leakage using chronological training, validation, and held-out test windows.
- Optimized threshold to 0.2138, reaching 0.9287 ROC-AUC and 0.5960 average precision.
- Served models through FastAPI with MLflow tracking, drift monitoring, Docker, and CI/CD.

Reward-Free Safe Reinforcement Learning - MSc Thesis

- Extended CEM and RENYI exploration algorithms with explicit safety constraints.
- Implemented PPO-style optimization, VAE density estimation, TD(lambda), and importance sampling.
- Evaluated coverage and safety across three environments and 24 fixed seeds.
- Found CEM with Shannon entropy achieved strongest feasible classic-control coverage.

Health Platform Text Classification - Bachelor End Project

- Analyzed text from 5,087 companies using reproducible double-coding procedures.
- Expanded 750 labels to 870 through 30 query-by-committee active-learning batches.
- Selected bag-of-words SVM achieving 0.610 F1 on independent test data.

EDUCATION

MSc Data Science and Artificial Intelligence | Eindhoven University of Technology

2023 - 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science | Eindhoven University of Technology and Tilburg University

2020 - 2023

- Bachelor End Project: Health Platform Text Classification Using Active Learning

Honors Academy - AI Track | Eindhoven University of Technology

2021 - 2023

- Projects: energy disaggregation, time-series modeling, visualization, and GANs

SKILLS

Programming: Python, SQL, R, C++, Bash

ML & Statistics: pandas, NumPy, scikit-learn, XGBoost, PyTorch, TensorFlow

Modeling: classification, NLP, time-series analysis, forecasting, feature engineering

Evaluation: cross-validation, temporal validation, imbalanced learning, threshold optimization

Analytics & MLOps: MLflow, FastAPI, process mining, KPI design, Plotly, Dash, Docker

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)